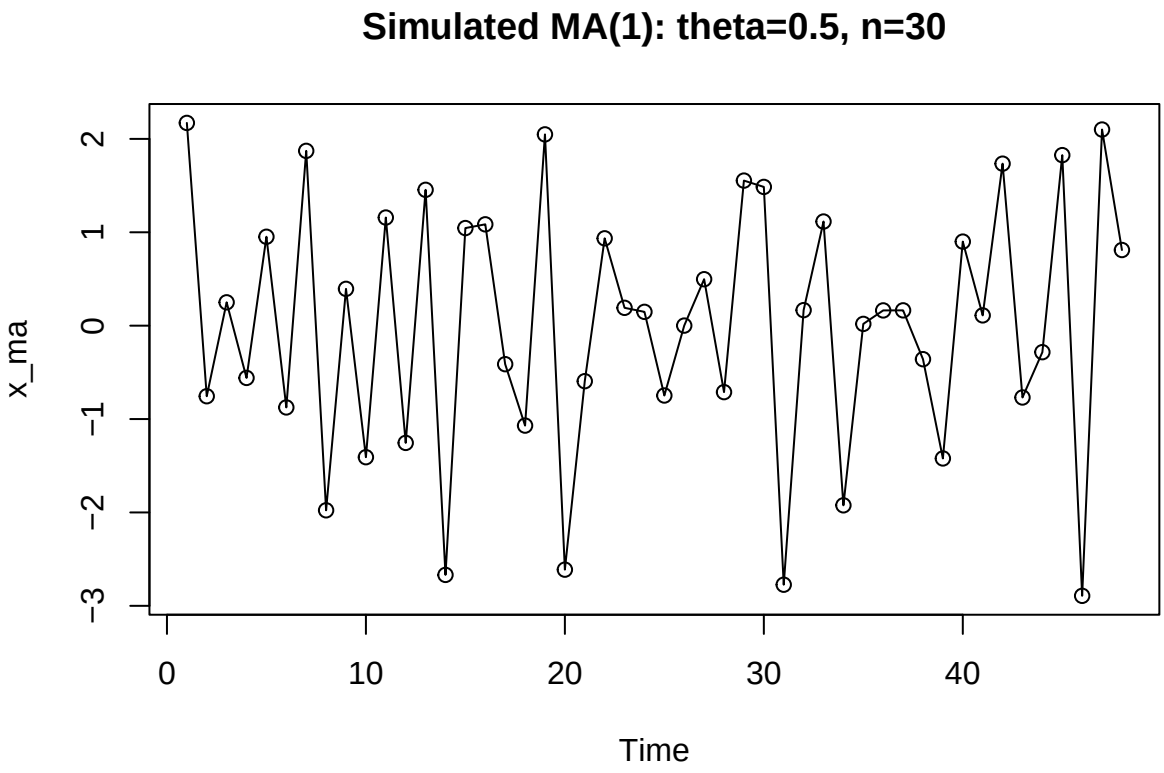
Exercise 8

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a)

```
n <- 48
theta_true <- 0.8
x_ma <- arima.sim(n = n, list(ma = -theta_true))
plot(x_ma, type = "o", main = "Simulated MA(1): theta=0.5, n=30")
```



b)

```
r1 <- acf(x_ma, lag.max = 1, plot = FALSE)$acf[2]
thetaMM1 = -1/(2*r1) + sqrt(1/(4*r1^2) - 1)  # solution 1
```

```
## Warning in sqrt(1/(4 * r1^2) - 1): NaNs produced
```

```
thetaMM2 = -1/(2*r1) - sqrt(1/(4*r1^2) - 1)  # solution 2
```

```
## Warning in sqrt(1/(4 * r1^2) - 1): NaNs produced
```

```
paste("Thetahat1: ", thetaMM1, "thetaMM2: ", thetaMM2)
```

```
## [1] "Thetahat1:  NaN thetaMM2:  NaN"
```

The Method of moment estimator of Theta is the solution with |theta| < 1 (invertible) which is thetaMM2 $= 0.686922977584318$

c)

```
fit_cls <- arima(x_ma, order = c(0, 0, 1), method = "CSS", include.mean = FALSE)
cat("MA(1) CLS Estimate:", -coef(fit_cls), "\n")
```

```
## MA(1) CLS Estimate: 0.6571597
```

d)

```
fit_mle <- arima(x_ma, order = c(0, 0, 1), method = "ML", include.mean = FALSE)
cat("MA(1) MLE Estimate:", -coef(fit_mle), "\n")
```

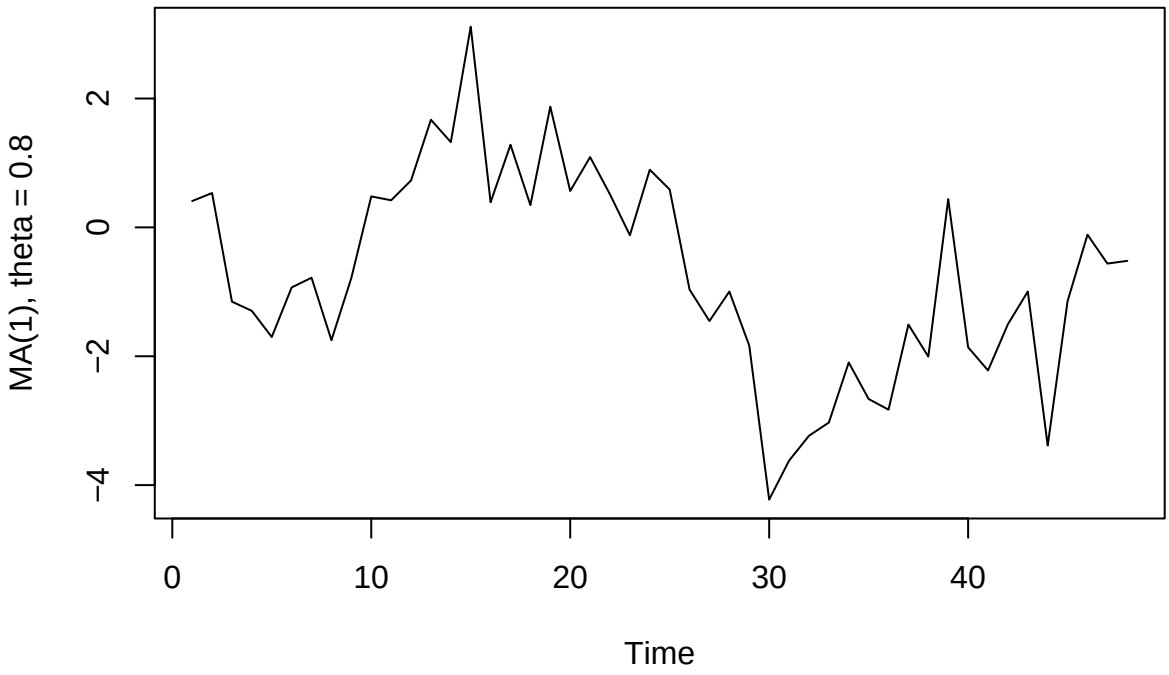
```
## MA(1) MLE Estimate: 0.6941318
```

e)

The method of moments estimator underestimates θ by 0.12, and the CSS and ML estimators overestimate θ by 0.1 and 0.13, respectively.
The best estimator remains the CLS estimator, which yields $\hat{\theta} = 0.91$.

f)

```
phi = 0.8
x_ar = arima.sim(model = list(ar = phi), n = n)
plot(x_ar, type="l", xlab="Time", ylab="MA(1), theta = 0.8")
```



Method of moment estimator of phi

```
phi_mom <- acf(x_ar, plot=FALSE)$acf[2]
cat("AR(1) MoM Estimate:", phi_mom, "\n")
```

```
## AR(1) MoM Estimate: 0.7259525
```

Least Square Estimation

```
fit_ar_cls <- arima(x_ar, order = c(1, 0, 0), method = "CSS", include.mean = FALSE)
cat("AR(1) CLS Estimate:", coef(fit_ar_cls), "\n")
```

```
## AR(1) CLS Estimate: 0.7780083
```

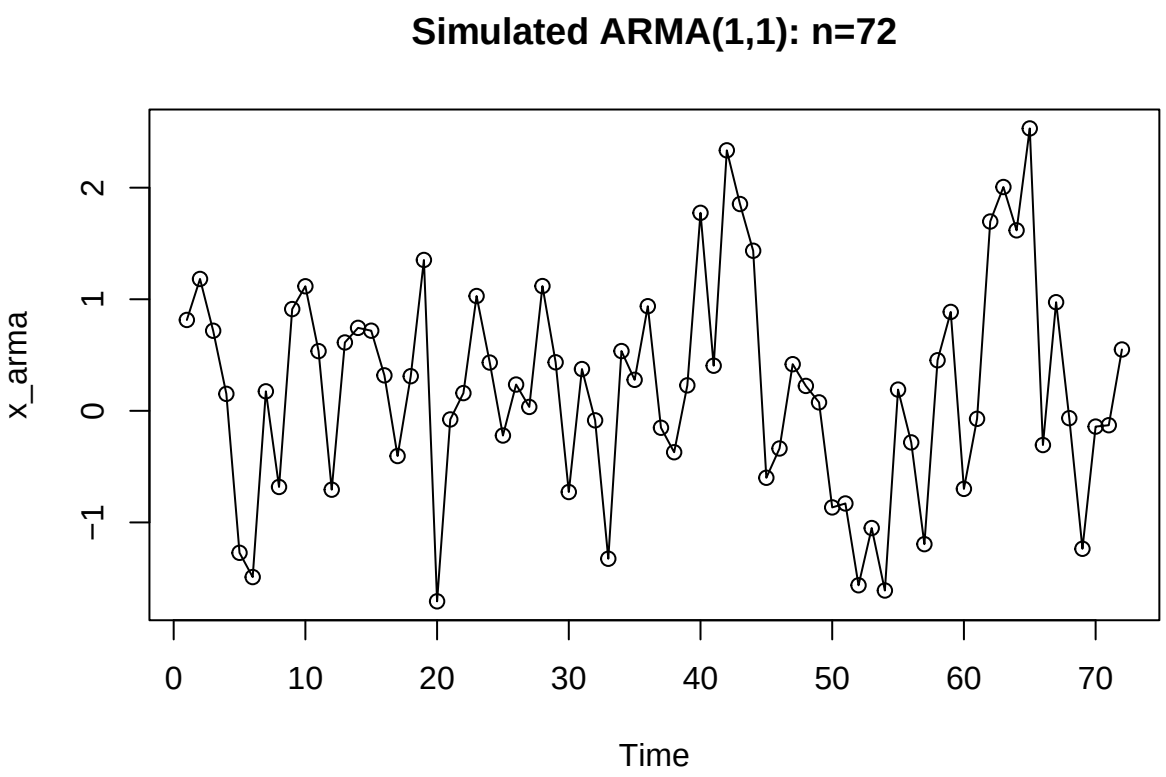
Maximum Likelihood Estimator

```
fit_ar_mle <- arima(x_ar, order = c(1, 0, 0), method = "ML", include.mean = FALSE)
cat("AR(1) MLE Estimate:", coef(fit_ar_mle), "\n")
```

```
## AR(1) MLE Estimate: 0.7637671
```

The best estimator of ϕ is the CSS estimator, which yields $\hat{\phi} = 0.77$.

```r
set.seed(103)
n <- 72
phi_t <- 0.7
theta_t <- 0.4
x_arma <- arima.sim(n = n, list(ar = phi_t, ma = -theta_t))
plot(x_arma, type = "o", main = "Simulated ARMA(1,1): n=72")
```




```r
r <- acf(x_arma, plot=FALSE)$acf[2:3]
phi_mom_arma <- r[2] / r[1]
cat("ARMA(1,1) MoM Estimate (phi):", phi_mom_arma, "\n")
```

```
## ARMA(1,1) MoM Estimate (phi): 0.4706208
```

```r
# (c) Conditional Least Squares (CLS)
fit_arma_cls <- arima(x_arma, order = c(1, 0, 1), method = "CSS", include.mean = FALSE)
print(coef(fit_arma_cls))
```

```
##        ar1        ma1
##  0.5699118 -0.2359007
```

```r
# (d) Maximum Likelihood (MLE)
fit_arma_mle <- arima(x_arma, order = c(1, 0, 1), method = "ML", include.mean = FALSE)
print(coef(fit_arma_mle))
```

```
##        ar1        ma1
##  0.5340801 -0.1865332
```

The best estimates of ϕ and θ are the CLS estimators, which yield: $\hat{\phi} = 0.56$ and $\hat{\theta} = 0.23$.